2024 Higher Chemistry Paper 2 – Q1

Section: Chemical Changes and Structure
Topic: Periodicity

Question summary (Q1):
Nitrogen and phosphorus are Group 5 elements.
(a) (i) Define “first ionisation energy”.
(a) (ii) Write the equation for the first ionisation of phosphorus.
(b) Explain why the melting point of phosphorus (P_4) is much higher than that of nitrogen (N_2), referring to intermolecular forces.

Worked Solution:
(a) (i) First ionisation energy = the energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous ions.

(a) (ii) $P(g) \rightarrow P^{+}(g) + e^{-}$

(b) • N_2 is a small diatomic molecule with only weak London dispersion forces between molecules.
• P_4 is a much larger tetrahedral molecule with many more electrons → stronger London dispersion forces between molecules.
• Stronger intermolecular forces mean more energy is needed to separate P_4 molecules, so phosphorus has a much higher melting point than nitrogen.

Final Answer:
(a) (i) Energy to remove one mole of electrons from one mole of gaseous atoms.
(a) (ii) $P(g) \rightarrow P^{+}(g) + e^{-}$
(b) Phosphorus has stronger London dispersion forces than nitrogen, so it has the higher melting point.

Revision Tips:

- First ionisation energy is always defined in terms of 1 mole, gaseous atoms, and removal of 1 electron.
- London dispersion forces increase with the number of electrons and molecular size.
- Small non-polar molecules (like N_2) have much weaker intermolecular forces than larger molecules (like P_4).